

SIMATS ENGINEERING ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA1710

COURSE OUTCOME COVERED

CO3: Analyze knowledge representation and reasoning using propositional logic, FOL, inference, resolution, chaining methods, knowledge representation to perform logical reasoning for drawing valid conclusions and decision making. (BL4)

Assessment Tool Weightage: 50%

Time Duration: 1 Hour

QUESTIONS: 4

Total Marks:40

Annexure A

CASE STUDY

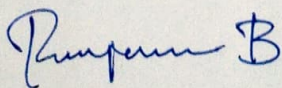
Code of Conduct:

(192411091)

I POOJASREE B (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



$25 + 25 + 19 + 18 + 8 = 95$



ARTIFICIAL INTELLIGENCE

CO3 - ATI

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CSA1710

1. Vehicle Fault Diagnosis System

a) Proposition Logic Representation

Let the following propositions represent the facts:

E = Engine Warning Light

N = Unusual Noise

O = Low Oil Pressure

H = Overheating Sign

F_1 = Possible Engine Failure Risk

F_2 = Possible Oil Leak

F_3 = Possible Coolant System Failure

The rules are:

1. Engine Warning Light AND Unusual Noise $\rightarrow$ Possible Failure Risk

$$(E \wedge N) \rightarrow F_1$$

2. Engine Warning Light AND Low Oil Pressure $\rightarrow$ Possible Oil Leak

$$(E \wedge O) \rightarrow F_2$$

3. Unusual Noise AND Overheating Sign $\rightarrow$ Possible Coolant System Failure

$$(N \wedge H) \rightarrow F_3$$

Facts about Vehicle X

$E = \text{True}$

$N = \text{True}$

$H = \text{True}$

There is no fact stating that Low Oil Pressure is True, so: $O = \text{Unknown}$

b) Forward Chaining

Forward chaining starts with the known facts and repeatedly applies rules whose conditions are satisfied.

Initial Facts

$E = \text{True}, N = \text{True}, H = \text{True}$

Step 1 - Apply Rule I

Rule 1: $(E \wedge N) \rightarrow F_1$

Since $E = \text{True}, N = \text{True}$

Therefore $F_1 = \text{True}$

~~Desired~~ fact - Possible Engine Failure Risk

Step 2 - Apply Rule II

Rule 2: $(E \wedge O) \rightarrow F_2$

We know: $E = \text{True}$ but: $O = \text{Unknown}$

So $F_2 = \text{Unknown}$

Possible Oil leak cannot be concluded

Step 3 - Apply Rule III

Rule IV: $(N \wedge H) \rightarrow F3$

We know: $N = \text{True}$, $H = \text{True}$

So, $F3 = \text{True}$

Derived fact: Possible Coolant System Failure

Therefore, the possible faults for Vehicle X are:

Possible Engine Failure Risk

$\&$
Possible Coolant System Failure

c) Backward Chaining

The goal is:

$F3 = \text{Possible Coolant System Failure}$

We work backward from the goal and find which rule can produce it.

Step 1: Start with Goal

Rule III gives: $(N \wedge H) \rightarrow F3$

$\therefore$ to prove $F3$, we need to prove:

$N \wedge H$

Step 2: Reduce the Goal

The goal becomes two sub-goals:

1. $N = \text{Unusual Noise}$

2. $H = \text{Overheating Sign}$

So : $F_3 \Leftarrow N \wedge H$

Step 3: Check the facts

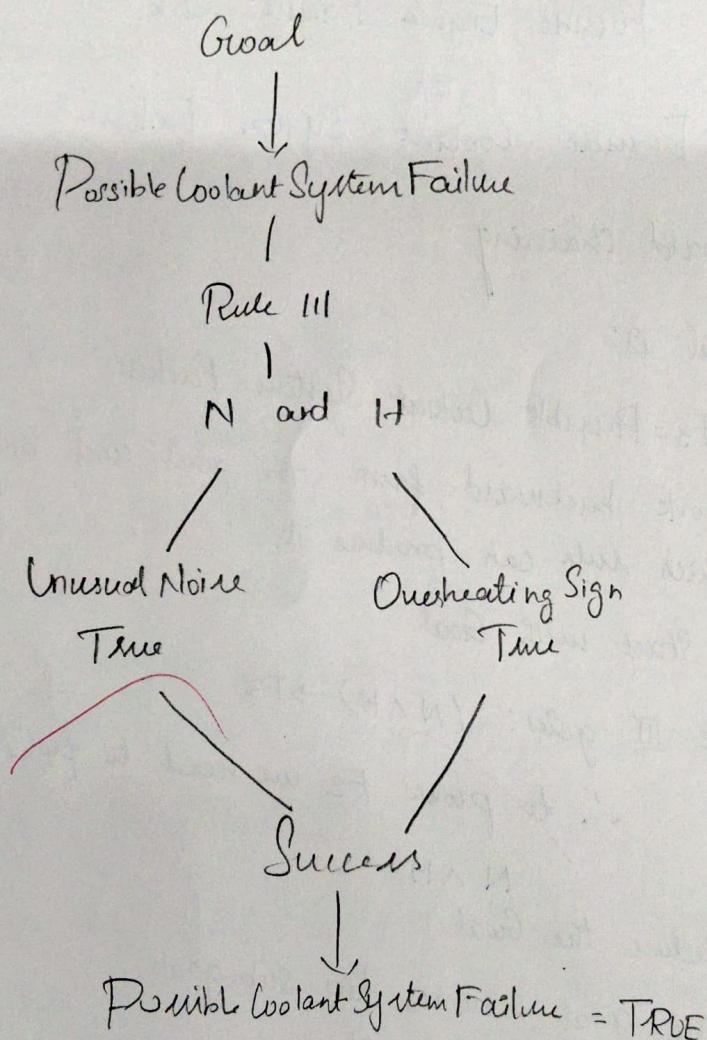
The knowledge base contains: $N = \text{True}$ & $H = \text{True}$

Therefore, both sub-goals are satisfied.

Hence: $N \wedge H = \text{True}$

$\Rightarrow$ Possible Coolant System Failure = True

Goal Reduction Tree



Hospital Emergency Ward Routing Agent

a) Unification

Rule 1

$\text{Patient}(x) \wedge \text{CriticalCondition}(x) \rightarrow \text{PriorityCorridor}(x)$

Given:

$\text{Patient}(\text{Patient A})$

$\text{CriticalCondition}(\text{Patient A})$

Match x with Patient A

$$\theta = \{x / \text{Patient A}\}$$

Therefore:

$\text{PriorityCorridor}(\text{Patient A})$

Unification Steps

1. $\text{Patient}(x)$ makes $\text{Patient}(\text{Patient A}) \rightarrow x = \text{Patient A}$
2. $\text{CriticalCondition}(x)$ matches $\text{CriticalCondition}(\text{Patient A})$
3. Conclusion $\rightarrow \text{PriorityCorridor}(\text{Patient A})$

b) Resolution Proof of Hold (Patient B)

CNF Clauses

Rule 1:

$\neg \text{Patient}(x) \vee \neg \text{CriticalCondition}(x) \vee \text{PriorityCorridor}(x)$

Rule 2:

$\neg \text{PriorityCorridor}(x) \vee \text{CorridorCleared}(x)$

Rule 3:

$\neg \text{Patient}(x) \vee \neg \text{Waiting Behind}(x, y) \vee \neg \text{Corridor Cleared}(y) \vee \text{Hold}(x)$

Negated goal:

$\neg \text{Hold}(\text{Patient B})$

Resolution steps

Step 1:

From Rule 1 + given facts:

$\text{Patient}(\text{Patient A}), \text{Critical Condition}(\text{Patient A})$

derive: $\text{Priority Corridor}(\text{Patient A})$

Step 2:

Using Rule 2:

$\text{Priority Corridor}(\text{Patient A})$

derive: $\text{Corridor Cleared}(\text{Patient A})$

Step 3:

Given: $\text{Patient}(\text{Patient B})$

$\text{Waiting Behind}(\text{Patient B}, \text{Patient A})$

$\& \text{Corridor Cleared}(\text{Patient A})$

Rule 3 gives: $\text{Hold}(\text{Patient B})$

Step 4:

We have both: $\text{Hold}(\text{Patient B})$

and the negated goal: $\neg \text{Hold}(\text{Patient B})$

Hence, $\text{Hold}(\text{Patient B})$ is proved true.

c) Two Simultaneous Critical Patients

The current knowledge base is not sufficient.

It only states:

CriticalCondition(Patient A)

It does not state that Patient B is critical
Required additional fact:

CriticalCondition(Patient B)

Additional rules are also needed for:

Corridor availability

Priority between two critical patients

Handling simultaneous routing

For example:

$CriticalCondition(n) \wedge Patient(n) \rightarrow Emergency$
 $Priority(n)$

Conclusion

The current KB can handle individual patients, but additional facts and priority / resource-allocation rules are required for two simultaneous critical patients.

3. Weather-Based Flight Scheduling Expert System

a) Convert into CNF

Let :

S = Storm Warning

D = Flight Delay Needed

I = International Route

N = Notify Passengers

C = Compliance Risk

$$P_1 \quad S \rightarrow D$$

Eliminate implication: $\neg S \vee D$

$$P_2 \quad D \wedge I \rightarrow N$$

$$\text{CNF: } \neg D \vee \neg I \vee N$$

$$P_3 \quad \neg N \vee \neg I \vee C$$

Facts S

I

b) Resolution Proof of Notify Passengers

Goal N

Negate the goal: $\neg N$

From P_1 and fact S : $\neg S \vee D$

Since S is true: D

From P_2 : $\neg D \vee \neg I \vee N$

Given: D, I

Resolution gives: N

Now we have both: N

and the negated goal: $\neg N$

Therefore: Notify Passengers is proved true.

Resolution Chain

$$S \rightarrow D$$

$$D \wedge I \rightarrow N$$

Therefore Notify Passengers = True

Remove Stormwarning

If Stormwarning is removed:

Step 1

$P_1: S \rightarrow D$

But S is no longer a fact

Therefore: D cannot be derived

Step 2

Without D , P_2 cannot fire.

Therefore: Notify Passengers cannot be derived

Step 3

P_3 requires: $\neg N \wedge I$

Although $I = \text{True}$, we only know that N cannot be derived. This does not mean $N = \text{False}$.
 $\therefore$ ComplianceRisk cannot be concluded.

Conclusion:

Removing ~~Stormwarning~~ breaks the chain $S \rightarrow D \rightarrow N$. ComplianceRisk still does not follow because absence of NotifyPassengers is not the same as $\neg \text{NotifyPassengers}$ in classical logic.

4. Cave Exploration Knowledge Agent

a) Belief State and Modus Ponens

Given percepts

Draft at $[1, 3]$

Glow at $[1, 1]$

Sparkle at $[2, 3]$

Apply rules

Rule 1:

$\text{Draft}[1, 3] \rightarrow \text{Pit-Adjacent}[1, 3]$

Since $\text{Draft}[1, 3]$ is true:

$\text{Pit-Adjacent}[1, 3]$

Rule 2:

$\text{Glow}[1, 1] \rightarrow \text{Lava Vent Adjacent}[1, 1]$

Therefore: $\text{Lava Vent Adjacent}[1, 1]$

Rule 3:

$\text{Sparkle}[x, y] \rightarrow \text{Treasure}[x, y]$

For $x=2, y=3$:

$\text{Treasure}[2, 3]$

Derived Facts

Pit is adjacent to $[1, 3]$

Lava vent is adjacent to $[1, 1]$

Treasure is at $[2, 3]$

2) Possible Locations of Pit and Lava Vent

Pit

From Draft at $[1,3]$, the pit is in a neighbouring cell.

Possible adjacent cells:

$[1,2], [1,4], [2,3]$

Lava Vent

From Glow at $[1,1]$, the Lava Vent is in a neighbouring cell.

Possible cells:

$[1,2], [2,1]$

Thus, $[1,2]$ is a possible location for both the Pit and Lava Vent.

c) Is the Path safe?

Path:

$[1,1] \rightarrow [1,3] \rightarrow [2,3]$

$[1,1]$: No Pit/LavaVent is directly known

$[1,3]$: Draft indicates a Pit is adjacent, but not necessarily at $[1,3]$

$[2,3]$: Treasure is present, but safety is not explicitly established.

$[1,2]$: It is a possible Pit/LavaVent location, so the route cannot be confirmed safe.

Rule 4 requires:

$$\neg \text{At}[x,y] \wedge \neg \text{LavaVent}[x,y] \rightarrow \text{Safe}[x,y]$$

But the KB does not provide enough information to prove these negative facts.

Conclusion

The Path cannot be guaranteed safe.

The agent knows the treasure is at [2,3], but it needs additional information to establish that every cell on the path is safe.